

Retail Customer Retention Analytics – WALMART

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Task 1: Data Modeling & Cleaning

1.1 Load and transform datasets in Power Query

Get Data-> Text/CSV->Select file-> Data type detection (based on entire dataset)-> Load

(same did for all 5 csv files)

1.2 Handle duplicates, missing values, and ensure correct data types

Select Transform->

Churn labelled customer table-> use first row as header->change data type from text to date of 'last purchase date' column

Customer demographics table-> use first row as header-> change data type from text to whole number of 'age' column -> change data type from text to date of 'membership since' column

Customer transaction table-> use first row as header-> change data type from text to date of 'transaction date' column-> change data type from text to decimal number of 'amount' column

Loyalty program table-> use first row as header-> change data type from text to whole number of 'points_earned' & 'points_redeemed' column

Store location table- > use first row as header-> change data type from text to whole number of 'opening_year' column

1.3 Create calculated columns:

- $\text{Membership_Duration} = \text{Today} - \text{Membership_Since}$

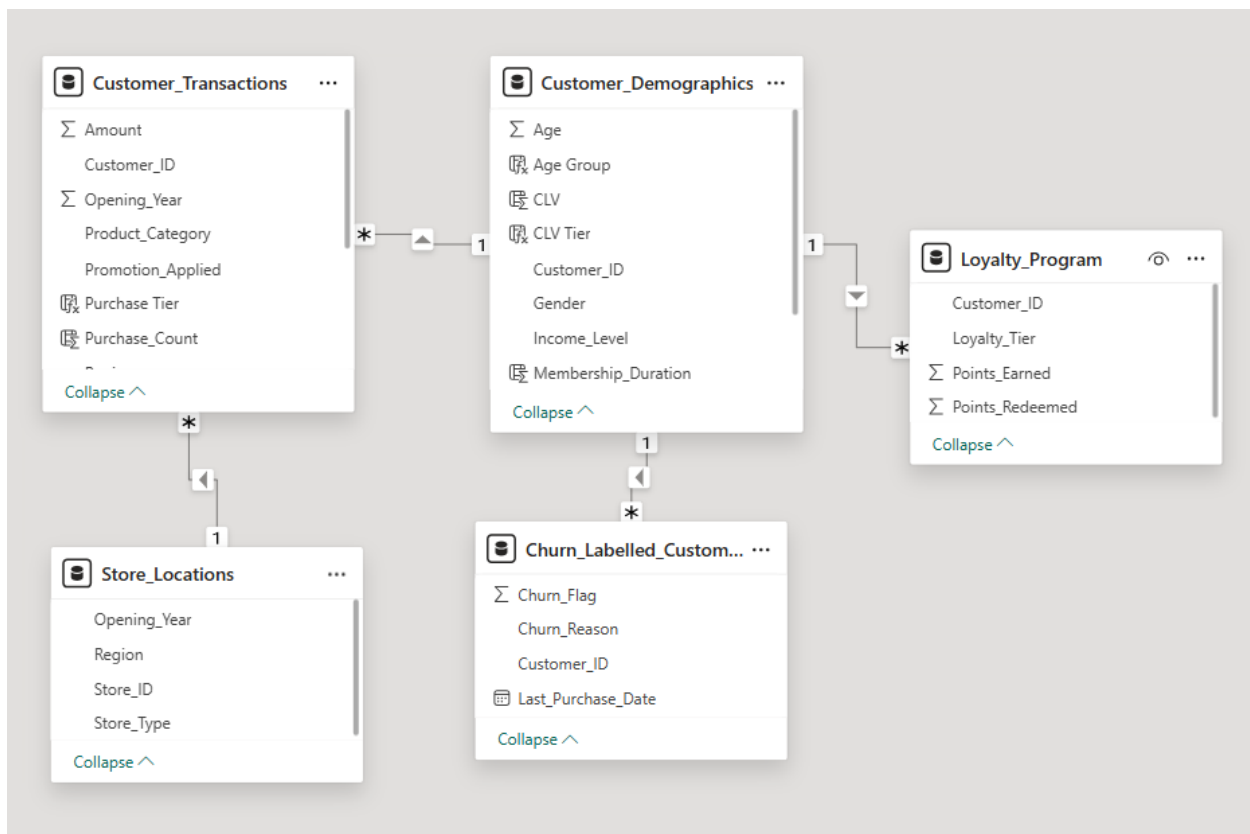
Add new calculated column in customer demographic table-> $\text{Membership_Duration} = \text{DATEDIFF}(\text{Customer_Demographics}[\text{Membership_Since}], \text{TODAY}(), \text{YEAR})$

- Extract Transaction_Year, Transaction_Month

Customer transaction table-> select 'transaction date' column-> add column-> date-> year-> name of the month-> rename them as transaction year and transaction month

1.4 Create a basic Data Model view

- One-to-Many: Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers
- Many-to-One: Transactions → Store_Locations



Task 2: Churn & Retention Metrics

2.1 Create **Churn Rate KPI** = (Churned Customers / Total Customers) * 100

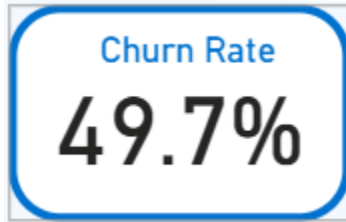
Create Measure=>

Total Customers = `DISTINCTCOUNT(Customer_Demographics[Customer_ID])`

Churned Customers =

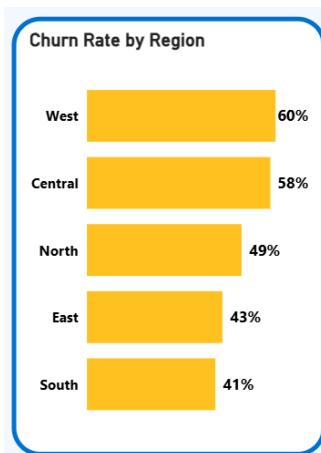
`CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]), Churn_Labelled_Customers[Churn_Flag]=1)`

Churn Rate = $\text{DIVIDE}([\text{Churned Customers}], [\text{Total Customers}])$

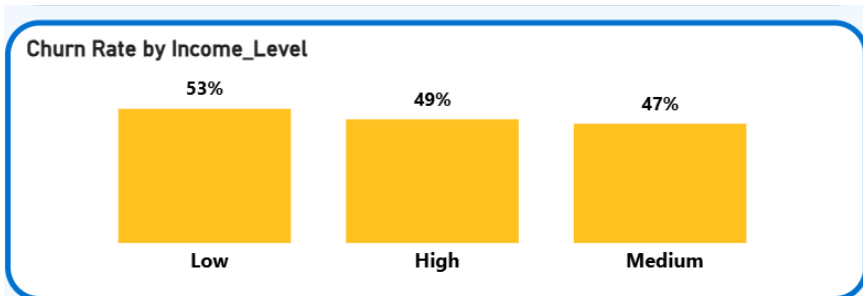


2.2 Visualize churn rate by:

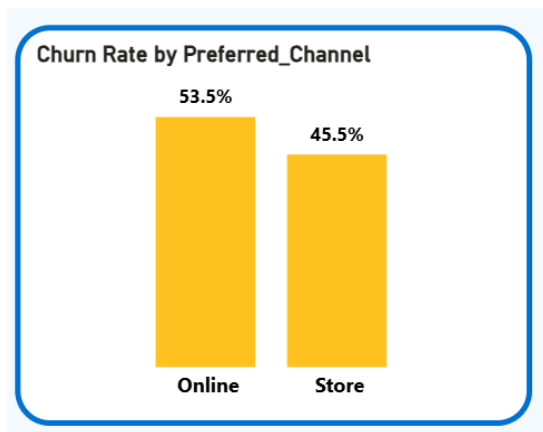
- **Region**



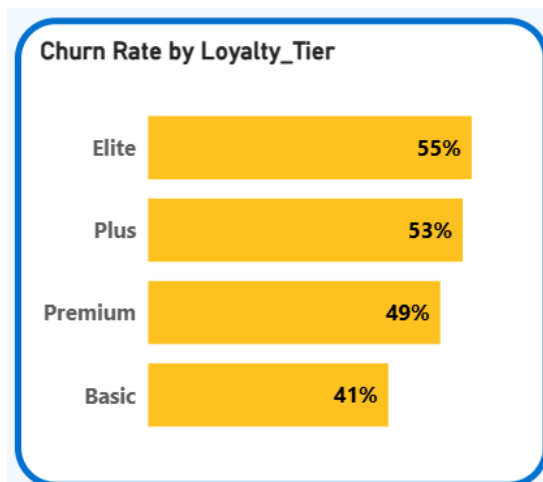
- **Income Group**



- **Channel (Store/Online)**



- **Loyalty Tier**



2.3 Funnel Chart: Total Customers → Repeat Customers → Churned

In customer transaction table, Add calculated column named as 'purchase_count' ->

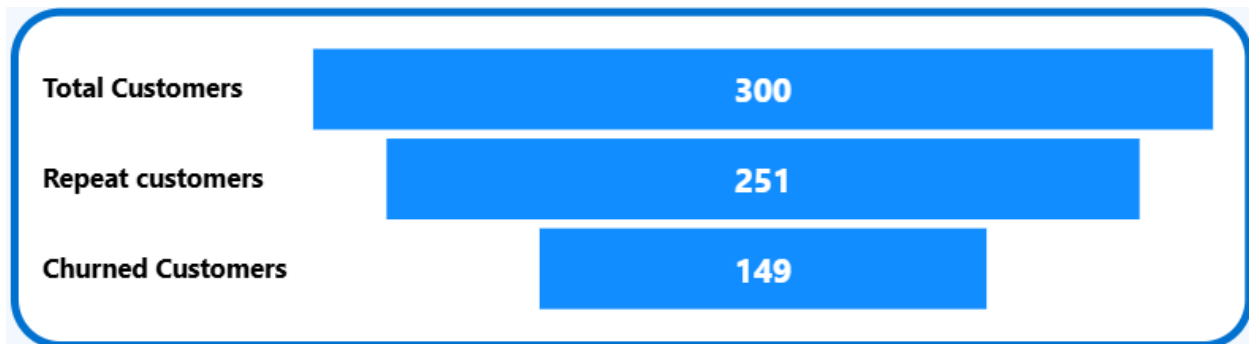
Purchase_Count =

`CALCULATE(COUNT('Customer_Transactions'[Transaction_ID]), ALLEXCEPT(Customer_Transactions, Customer_Transactions[Customer_ID]))`

Add Measure ->

Repeat customers =

CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),
Customer_Transactions[Purchase_Count]>1)



Task 3: Repeat Purchase Analysis

3.1 Segment customers:

- **Low-Tier:** 0–3 purchases
- **Mid-Tier:** 4–8 purchases
- **High-Tier:** 9+ purchases

In Customer transaction table, add calculated column->

Purchase Tier = IF(Customer_Transactions[Purchase_Count]<=3,"Low-Tier",IF(Customer_Transactions[Purchase_Count]<=8,"Mid-Tier","High-Tier"))

3.2 Compare avg. purchase frequency by **Region, Age Group, Loyalty Tier**

Create Measure->

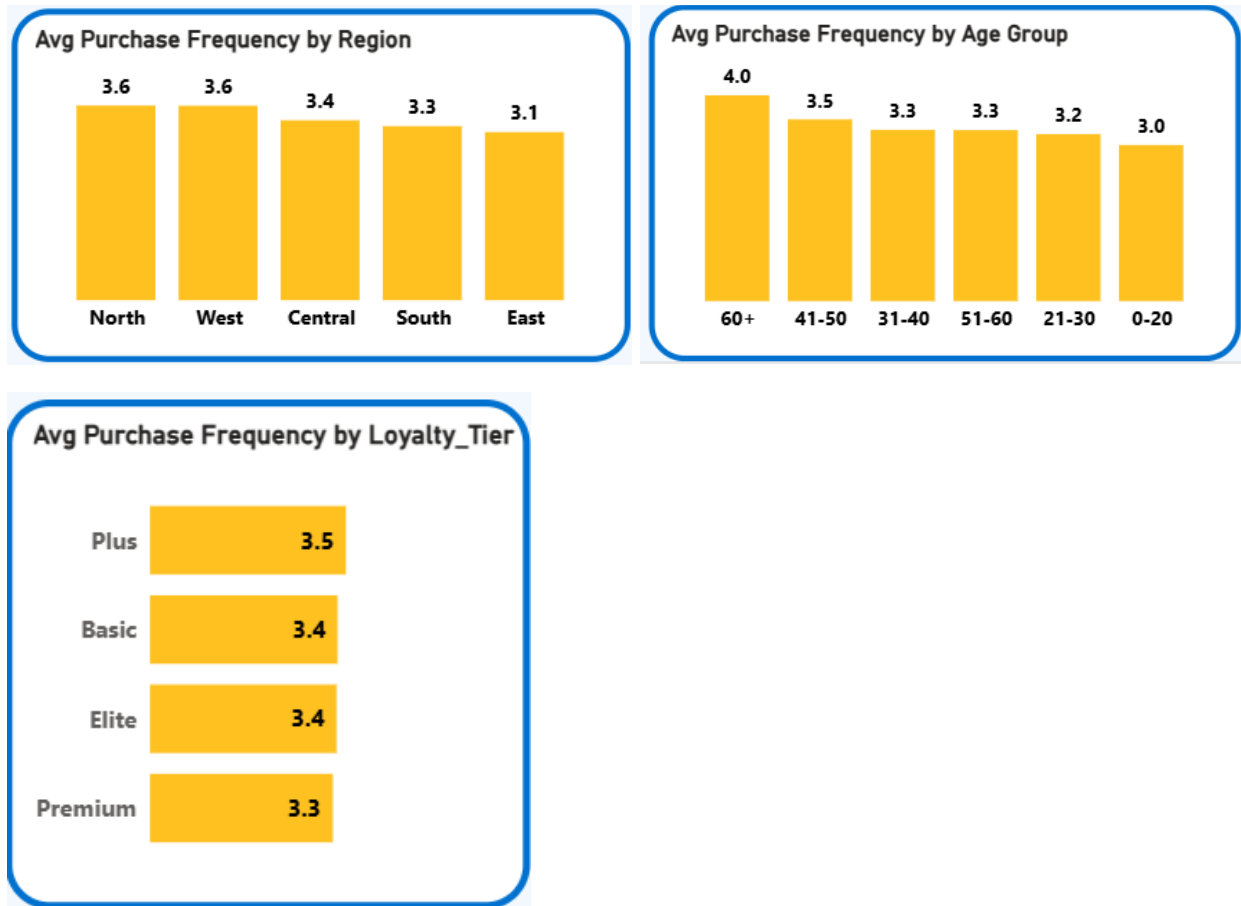
Avg Purchase Frequency =

DIVIDE(COUNTROWS(Customer_Transactions),DISTINCTCOUNT(Customer_Transactions[Customer_ID]))

Add Calculated column in Customer Demographics table->

Age Group = SWITCH(TRUE(),Customer_Demographics[Age]<=20,"0-20",Customer_Demographics[Age]<=30,"21-

30",Customer_Demographics[Age]<=40,"31-40",Customer_Demographics[Age]<=50,"41-50",Customer_Demographics[Age]<=60,"51-60",Customer_Demographics[Age]>60,"60+")



3.3 Identify **most purchased product categories by loyal customers**

Let's consider High-Tier & Mid-Tier customers as loyal customers, so add clustered column chart->

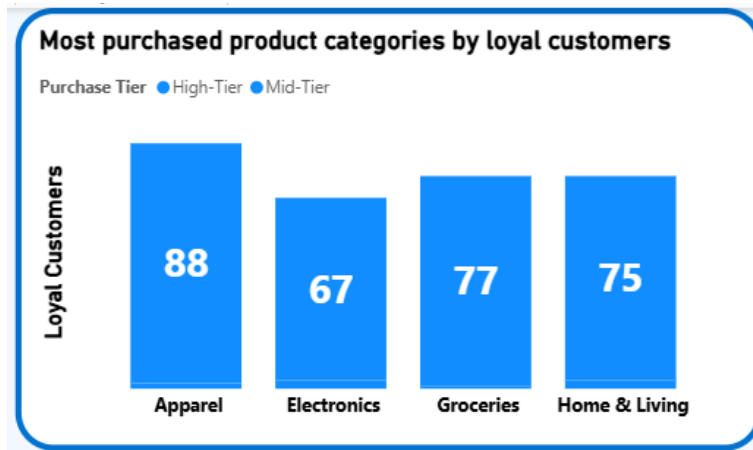
X-axis-> Product Category, Churn Flag=0

Y-axis-> count of Transaction_ID

Legend-> Purchase Tier

In filter pane, selected Mid & High-Tier only in basic filtering

In filter pane, select churn flag = 0 (as active customers)



Task 4: Promotion & Loyalty Impact

4.1 % of transactions with promotion applied

Calculate Measure->

Total Transactions = `DISTINCTCOUNT(Customer_Transactions[Transaction_ID])`

Transactions with Promotion Applied =

`CALCULATE(DISTINCTCOUNT(Customer_Transactions[Transaction_ID]),Customer_Transactions[Promotion_Applied]="Yes")`

Promotion Applied % = `DIVIDE([Transactions with Promotion Applied],[Total Transactions])`



4.2 Compare **avg. purchase amount with vs without promotions**

Add measure->

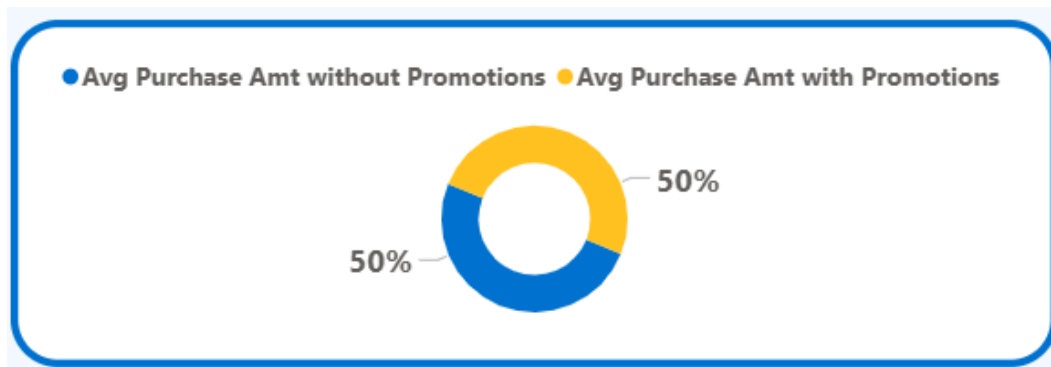
Avg Purchase Amt with Promotions =

`CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied]="Yes")`

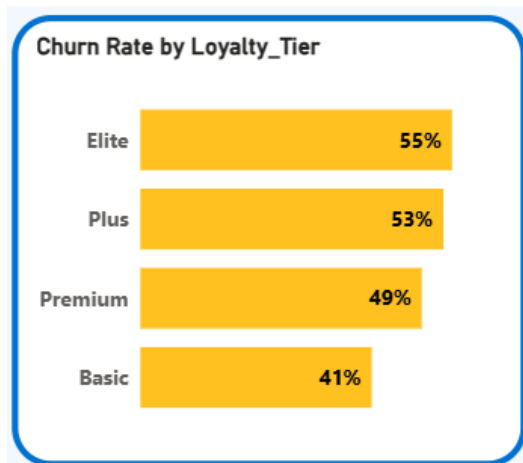
Avg Purchase Amt without Promotions =

`CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied]="No")`

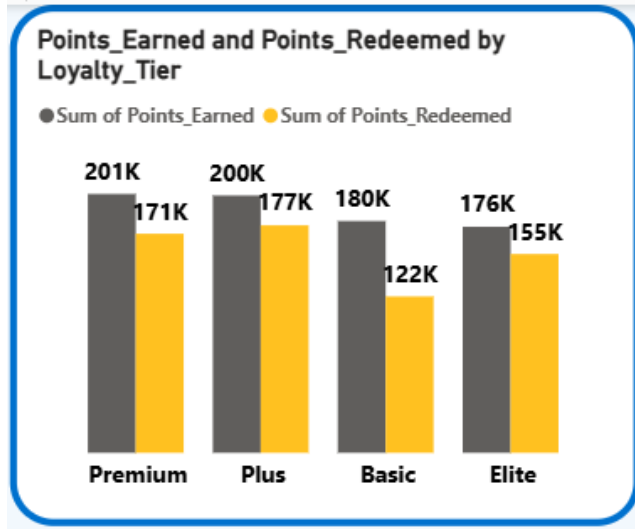
Add a donut chart to do the comparison



4.3 Churn rate across **loyalty tiers**



4.4 Points Earned vs Redeemed by Tier (clustered column chart)



4.5 Recommendations to **improve redemption & retention**

Make Redemption Easier & More Visible

- Auto-apply points at checkout (online & in-store) as default.
- Add a permanent “Use Points” toggle on Walmart.com and app checkout
- Push notification & email: “You have 18,500 points (\$185–\$250 value for Elite) – use them before they feel worthless!”.

Task 5: Store & Channel Performance vs Retention

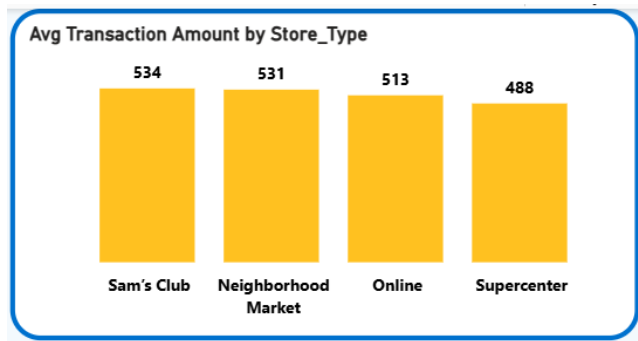
5.1 Merge store data with transactions

Transform-> Select Customer Transaction table -> Merge queries-> Select ‘Store location’ table-> Join Kind: Left outer-> OK

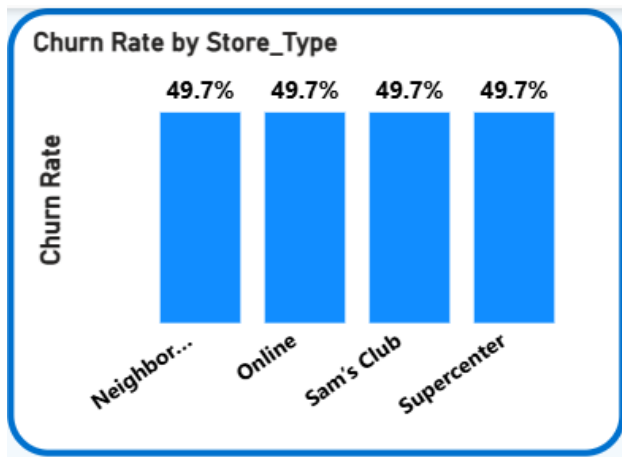
5.2 Visualize:

- Avg. transaction amount by **Store Type**

Take average of Amount on X-axis



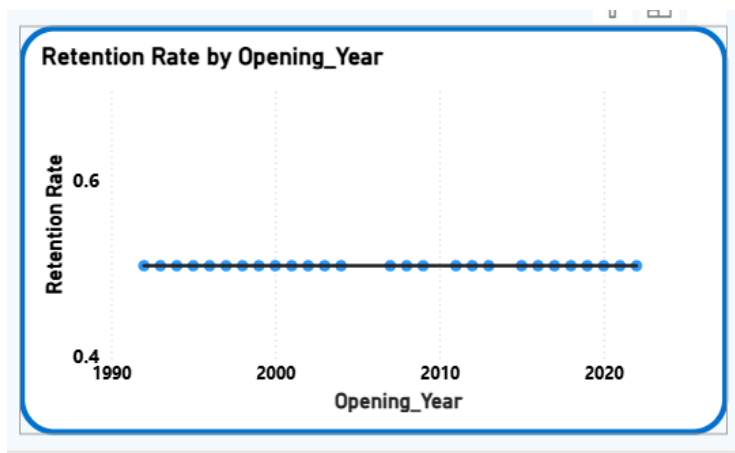
- Churn rate by store type



- Correlation between **store opening year & retention**

Add measure->

Retention Rate = 1-[Churn Rate]



No Correlation found between Retention rate and Opening year

Task 6: Customer Lifetime Value (CLV) Analysis

- $CLV = \text{Total Amount Spent} / \text{Membership Duration (Years)}$

Add calculated column in Customer demographics table->

Total Amount Spent =

`CALCULATE(SUM(Customer_Transactions[Amount]),ALLEXCEPT(Customer_Demographics,Customer_Demographics[Customer_ID]))`

Add calculated column in Customer demographics table->

$CLV = \text{DIVIDE}([\text{Total Amount Spent}], \text{Customer_Demographics}[\text{Membership_Duration}], 0)$

$CLV = \text{DIVIDE}([\text{Total Amount Spent}], \text{Customer_Demographics}[\text{Membership_Duration}], 0)$



- Segment customers into **Low, High CLV**

- Above Average CLV as High
- Below Average CLV value- Low

Add calculated column in Customer demographics table->

CLV Tier =

`IF(Customer_Demographics[CLV]>AVERAGE(Customer_Demographics[CLV]),"High","Low")`

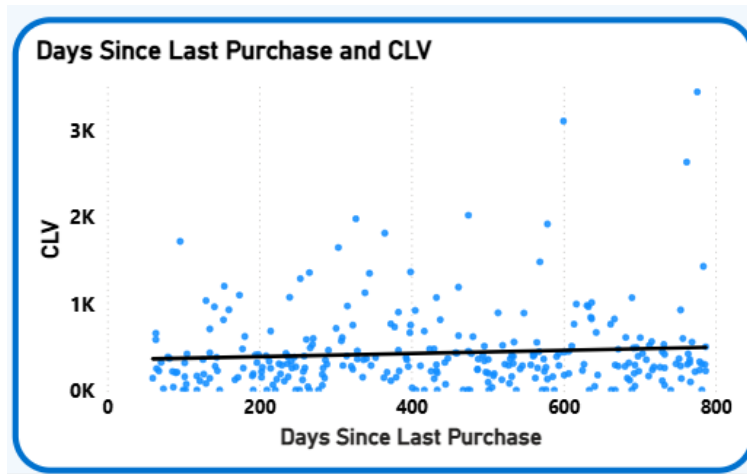
- Visualize:

- CLV vs Days Since Last Purchase

Add calculated column in Churn Labelled customers->

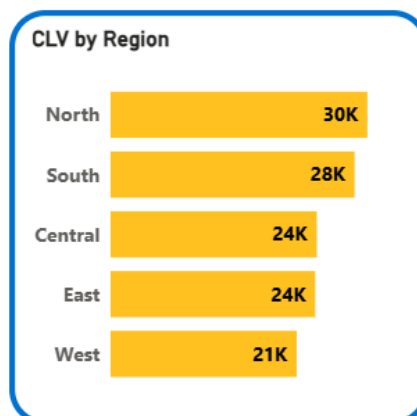
Days Since Last Purchase =

`DATEDIFF(Churn_Labelled_Customers[Last_Purchase_Date],TODAY(),DAY)`



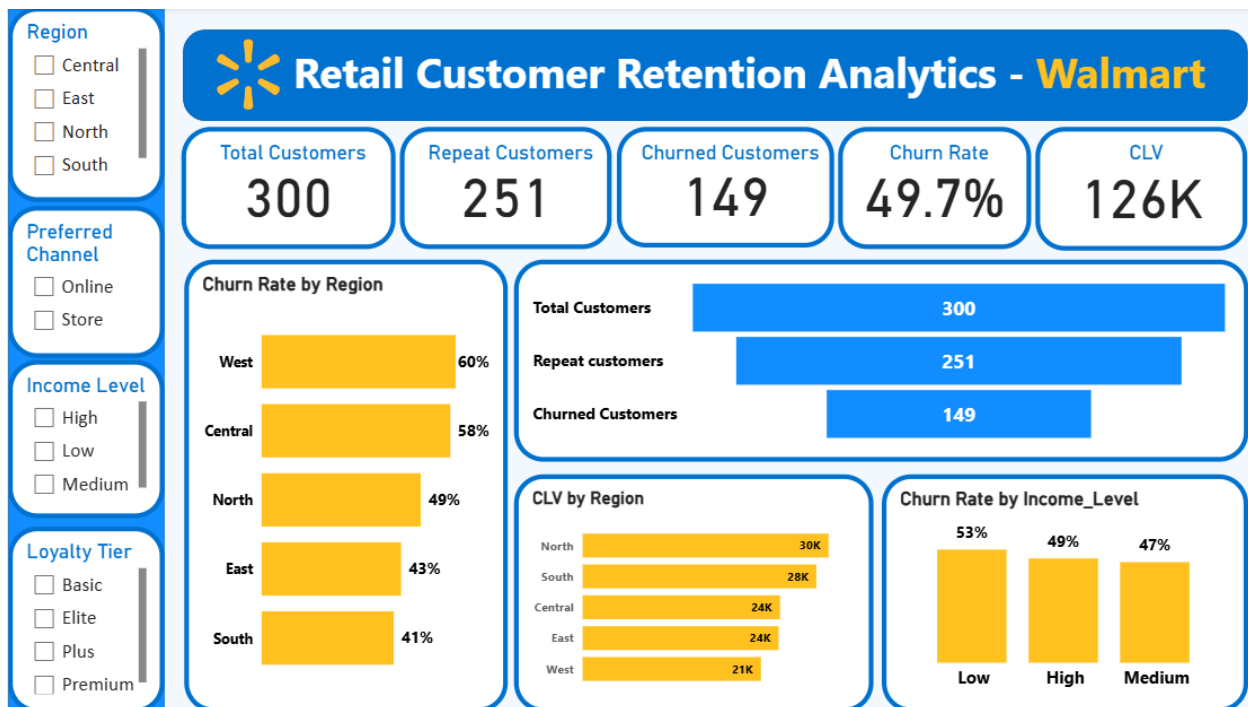
Found no significant correlation between CLV & Days since last purchase.

- CLV by Loyalty Tier & Region

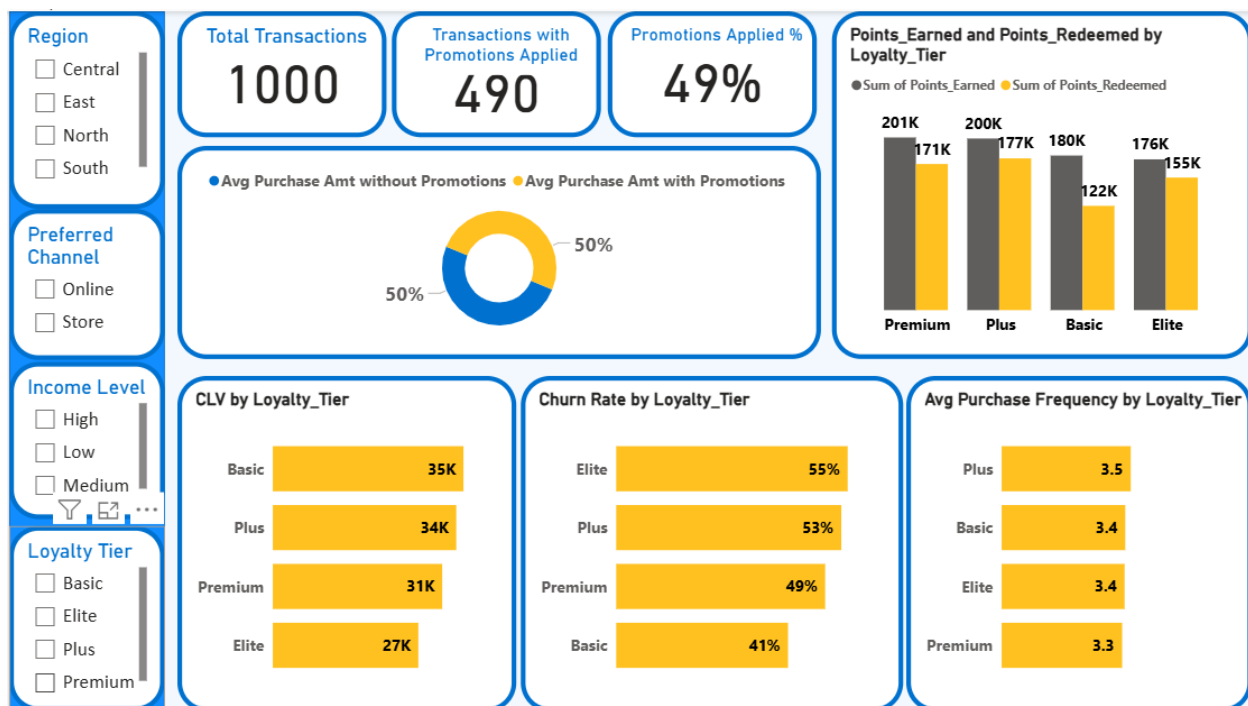


Task 7: Final Dashboard & Executive Summary

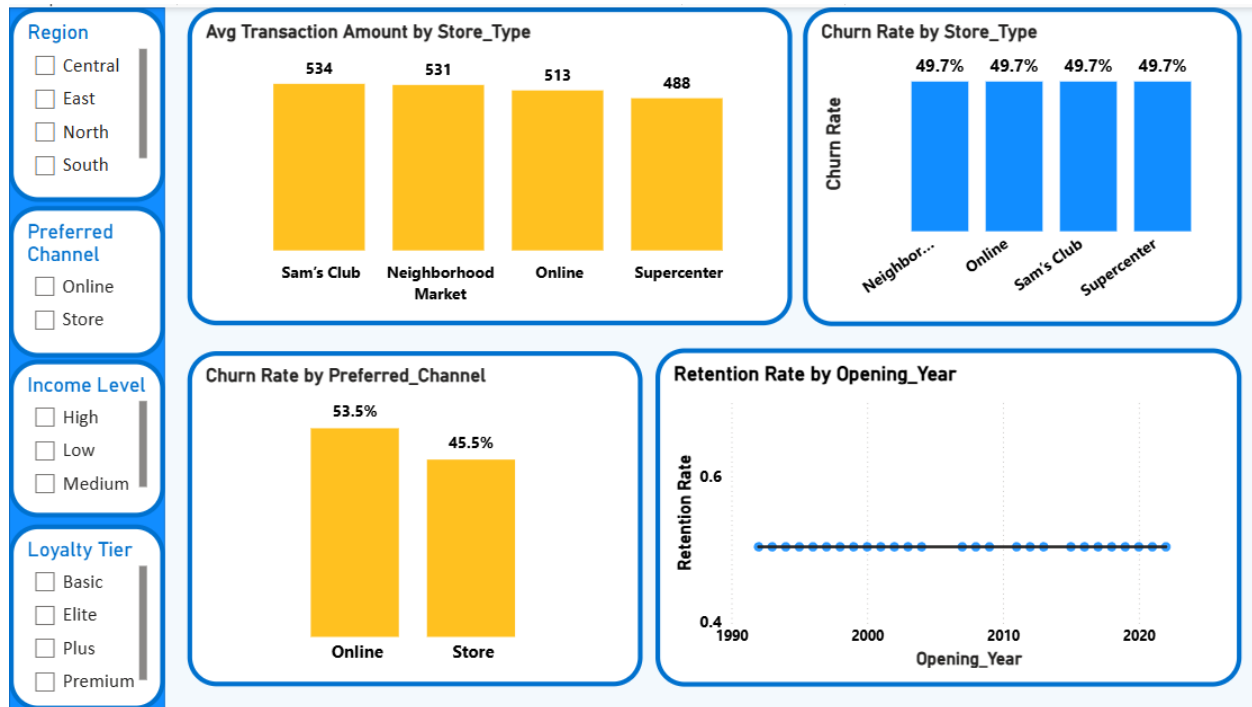
- Multi-page Power BI Report:
- Page 1: KPIs (Churn, CLV, Repeat Rate)



- Page 2: Loyalty & Promotion Impact

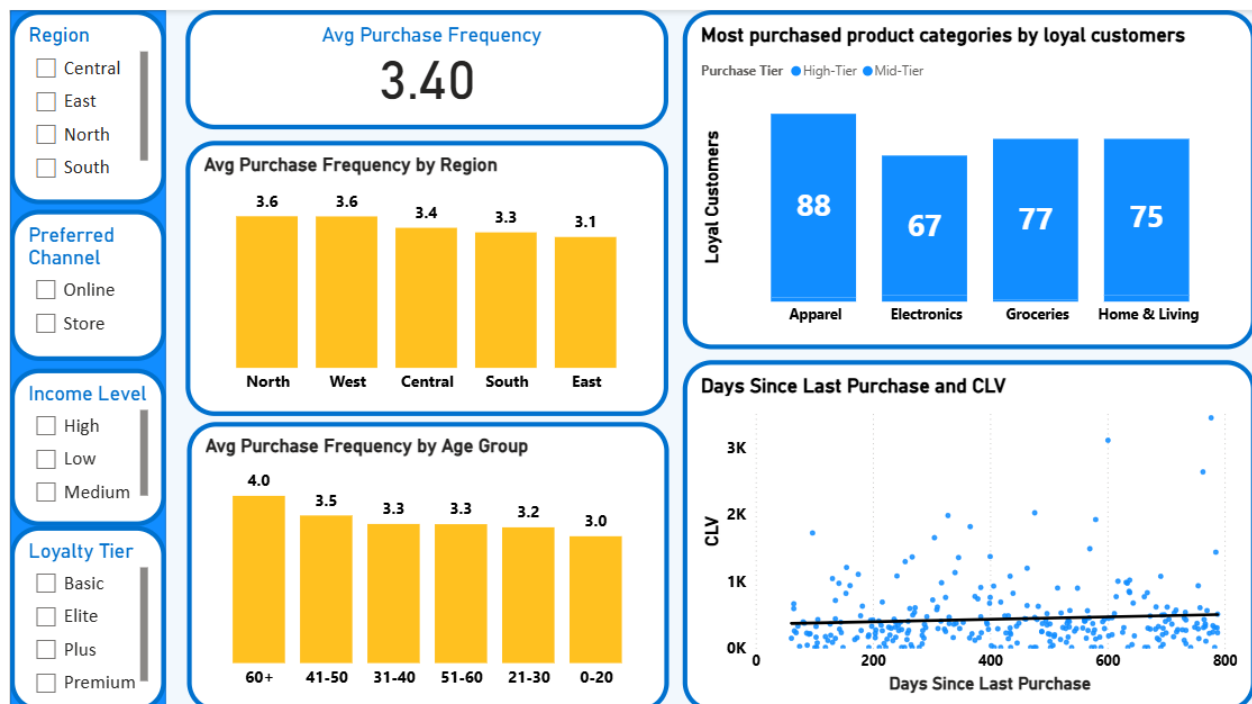


- Page 3: Store/Channel Insights



○ Page 4: Segmentation (Churned, Repeat, High-Value)

● Slicers: Region, Channel, Income, Loyalty Tier



● Top 3 recommendations for Walmart :

1. Which customers to prioritize for retention?

Priority 1: Elite & Plus loyalty tier members → Currently have the highest churn (55% and 53%) despite being "top" tiers. → These customers should be the most loyal but are leaving fastest.

Priority 2: West region customers → 60% churn (worst region) and lowest CLV (21K).

Priority 3: Low-income customers → 53% churn.

2. Which channels are underperforming?

- Physical stores (especially Supercenters)
- Store channel churn: **53.5%** vs Online: **45.5%** → Online retains 8% points better.
- Supercenter has the lowest average transaction value (\$488).
- All physical store types have identical ~49.7% churn → no format is winning.

3. How to strengthen loyalty program engagement?

The current loyalty program is upside-down:

- Basic tier has lowest churn (41%) and highest CLV (35K)
- Elite tier has highest churn (55%) and lowest CLV (27K)

Possible Root causes:

- Higher tiers earn many points but redeem very few → perceived low value of points.
- No clear incremental benefit for moving up tiers.

Action Plan to fix loyalty program:

1. Dramatically improve point redemption experience.
2. Make points easier to redeem online and in-store → Introduce instant redemption at checkout → Add high-value exclusive rewards (early access, free shipping, etc.)
3. Re-structure tier benefits → Give Elite/Plus members truly valuable perks (e.g., free unlimited delivery, 5-10% cashback, exclusive events) → Current benefits are clearly not compelling enough.

VIDEO LINK:

<https://screenrec.com/share/oH7anr5eds>